

Protocol for DNA extraction from whole blood using simple salting out procedure

Purpose: High-molecular weight DNA extraction from whole blood for genetic analyses, including no amplification long-range sequencing.

Written by: Birgitt Schuele

References: Miller et al 1988, DOI:10.1093/nar/16.3.1215

BUFFERS AND SOLUTION RECIPES

Reagent	Cat.No.	Purity	Vendor	Amount
Ammonium chloride, NH ₄ Cl	A9434	Molecular Biology, >99.5%	Sigma	500g
Potassium bicarbonate, KHCO ₃	60339	BioUltra, >99.5%	Sigma	500g
EDTA, 0.5M Solution	324504	ULTROL Grade	Millipore	500ml
Tris-HCl/Trizma	T5941	Bioperformance, >99%	Sigma	100 or 500g
Sodium chloride	S3014	Molecular biology, DNase, RNase & protease, none detected, ≥99%	Sigma	500g or 1kg
200proof absolute ethanol	EX0276	GR ACS	Supelco/ Millipore	1000ml
Isopropanol/2-propanol	278475	anhydrous, 99.5%	Sigma	1000ml
Proteinase K	P2308	lyophilized powder, BioUltra, ≥30 units/mg protein, for molecular biology	Sigma	25mg or larger amount

Note: If using different vendors, choose molecular biology grade reagents.

10X Red blood cell (RBC) lysis buffer (1L)

NH₄Cl 82.91g

KHCO₃ 10.01g

EDTA 0.5M 20ml

*complete to 1000ml with autoclaved dH₂O

*dilute to 1x before use with autoclaved dH₂O

10X Nucleus lysis buffer (100ml)

NaCl 23.38g

Tris-HCl 1M pH=8 10ml

EDTA 0.5M 4ml

*complete to 100ml with autoclaved ddH₂O

*dilute to 1x before use with autoclaved ddH₂O

20% SDS (used end of Day 1 for DNA extraction to poke holes in cell membrane)

20g SDS

80mL with autoclaved ddH₂O

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5M NaCl

146.1g NaCl

Bring volume to 500mL with autoclaved dH₂O

70% EtOH (used Day 2 of DNA extraction to “clean” DNA)

35mL 100% EtOH

15mL ddH₂O

TE (used at end of DNA extraction to re-suspend DNA and prepare for storage)

1mL of 1M Tris-HCl

0.2mL of 0.5M EDTA

Bring volume to 100mL with autoclaved ddH₂O

1M Tris-HCl, pH 8 (used to make nucleolysis buffer and TE)

15.7g Tris-HCl

Bring volume to ~95mL with autoclaved ddH₂O

→ measure & adjust pH to 8

Bring *final volume* up to 100mL with autoclaved ddH₂O

Proteinase K reconstitution:

Solutions can be prepared in 25 mM Tris-HCl buffer, pH 8.0, containing 1 mM Calcium chloride. Amounts for typical usage is 50-200 µg/mL. Solutions are stable at pH 8 at 4°C. It is stable in a broad range of environments: pH, buffer salts, detergents (SDS), and temperature. In the presence of 0.1-0.5% SDS, proteinase K retains activity and will digest a variety of proteins and nucleases in DNA preparations without compromising the integrity of the isolated DNA.

Materials:

1. 10 ml EDTA collection tube (K₂ EDTA, 5.4mg; 16 x 100mm; 10mL; Pink)
2. 50 ml conical tube (Fisher catalog # **12565268**)
3. 1.5ml clear Eppendorf tube (Fisher catalog # **5408129**)
4. 1000ul filter pipette tips
5. 5.0 ml serological pipette (Fisher catalog # **13-678-11D**)
6. 50 ml serological pipette (Fisher catalog # **13-678-11F**)
7. Gloves
8. Bleach (e.g. Clorox) as 20% fresh solution)

Equipment:

1. Centrifuge: Sorvall Legend XTR, or SorvallIT 6000B, H10000B
2. Centrifuge: Eppendorf 5417R
3. Rotator

Storage Vial labeling:

3 lines, 6pt bold font; 1.5” label

- a) Subject ID#: four digit SUSL ID (e.g. SUSL-2345), Sample type: *WBC*
- b) Date of blood draw: MM/DD/YY
- c) Study: [Name Study], Initials

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DNA extraction from human blood

Red blood cell lysis:

IF FROZEN blood:

1. In the morning take out tubes of blood from the -80C freezer. Record ID numbers and put in fridge (4 degrees) to thaw until later in the afternoon.
2. Once blood has thawed, invert several times or place on rotator for a couple of minutes and transfer to 50ml conical tubes. Continue to section: **Red blood cell lysis, if fresh blood.**

IF FRESH blood:

1. Decant the blood sample into 50 ml conical tube.
2. Add 1x RBC lysis buffer up to 50ml.
3. Place on shaker at room temperature for 30min (Note: blood solution becomes transparent).
4. Centrifuge 15min at 2000rpm at room temperature.
5. Discard supernatant in container with 20% bleach. (Note: The pellet contains the nuclei.)
6. Add 700ul 1x RBC lysis buffer and close the tube tightly. Pipette up and down to resuspend the pellet (cleans up the nuclei pellet from additional RBCs and allows for storage in small tubes).
7. Transfer to 1.5ml pre-labeled clear Eppendorf tube.
8. Centrifuge 5min at 2000rpm at room temperature.
9. Using a 1000 ul pipette tip, discard the supernatant in container with 20% bleach.
10. Pellets can be frozen at this point at -80C.

If direct processing with **Nuclei lysis, day 1**, then continue after step 5 of this section with **Nuclei lysis**, step 1.

Nuclei lysis (Day 1, afternoon)

1. Add 3ml of Nucleus Lysis Buffer and close tube tightly.
2. Pipette up and down to resuspend pellet.
3. Add 300ul of 10% SDS and 70ul Proteinase K (10mg/ml).
4. Mix by gently swirling tube.
5. Incubate overnight in water bath at 55 degrees.

DNA extraction (Day 2, morning):

1. Add 1ml of 5M NaCl to the tubes.
2. Close caps tightly and shake vigorously. Note: Very important to shake thoroughly at this step to precipitate proteins.
3. Centrifuge for 20min at 3000rpm at room temperature.
4. While centrifuging, label 2x 15ml conical tubes *per sample*.
5. Transfer supernatant to a 15ml conical tube and then repeat centrifugation for 20min at 3000rpm (Note: *The supernatant contains the DNA.*)
6. Transfer supernatant to next 15ml conical tube.
7. Add equal volume of ice-cold isopropanol (usually about 4ml).
8. Invert tubes several times and DNA will precipitate. (Note: *You should be able to see white strings (this is the DNA!)*)

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9. Centrifuge for 5min at 3000rpm.
10. Discard supernatant.
11. Add 1ml 70% ethanol to each tube.
12. Centrifuge for 5min at 3000rpm.
13. Carefully pipette off ethanol to not dislodge the DNA pellet.
14. Leave DNA to dry in uncapped tubes overnight.

Dissolve DNA

1. Add 250 μ l TE and allow DNA to dissolve in 15ml conical tube stored in fridge (4 degrees).
2. Once completely dissolved, transfer to labeled sterile Eppendorf tube.
3. Label top with subject ID. Label side with date of extraction and concentration.
4. Spec all samples using Nanodrop. Dilute samples to stock at 250ng/ μ l-350ng/ μ l and to working dilution at 10ng/ μ l
5. Transfer samples into barcode tubes and log into DNA bank.